

# **Characteristics of the Mississippi Sound Estuary and Watershed**

Mississippi Sound Estuary Program

November 6, 2025

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# Preface

This website is a collection of documents characterizing the Mississippi Sound Watershed, primarily through use of geospatial data. The outputs are compiled as a ‘[quarto book](#)’. Each chapter is produced by a separate `.qmd` file.

The purpose of the book format is to make the outputs readily findable and accessible. It may also serve as an outline for future undertakings like the State of the Sound report.

**Part I**

**Geographic Setting**

Some info here about different ways of describing where we are and what comprises the watershed: USGS HUCs, DEQ's division into different basins, and EPA EcoRegions.

Possibly also the definitions from the CCMP of MS Sound, MS Sound Estuary, and MS Sound Watershed.

# 1 Watersheds

## 1.1 HUCs

Nice explanation of HUCs on [this USGS page](#).

The HUCs that make up the Mississippi Sound Watershed are 0317 (Pascagoula) and 0318 (Pearl).

Show some mappy breakdowns.

## 1.2 MDEQ Basin Boundaries

There are several waterways that don't actually end up in either the Pearl or Pascagoula Rivers. So the Mississippi Department of Environmental Quality uses a slightly different division.

NEED TO CONFIRM, but I think: HUC12s are the same between MDEQ's classification and the USGS HUCs. They're just grouped differently, and the streams that empty into bays or straight into the Sound are the 'Coastal Streams Basin'. There is also a 'Coastal Offshore' grouping, that is comprised of the Sound itself, the islands, and the 3-mile zone beyond the islands.

## About

The `.qmd` file that generated this section was: `watersheds.qmd`.

For information on data processing, see Appendix [A.1.1](#).

## 2 EPA Ecoregions

For some geological classification that isn't based on watershed boundaries, we turn to the [EPA's Ecoregions](#).

### 2.1 Level 3 Ecoregions

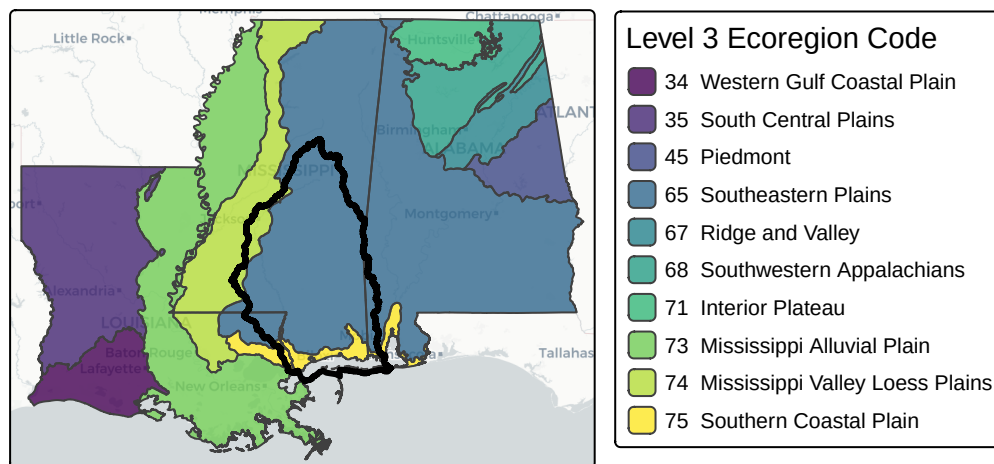


Figure 2.1: Level 3 Ecoregions in Louisiana, Mississippi, and Alabama; the Mississippi Sound Watershed is outlined in black.

The Mississippi Sound Watershed contains 4 level-3 ecoregions:

- 65 Southeastern Plains
- 73 Mississippi Alluvial Plain (in the very southwest corner)

- 74 Mississippi Valley Loess Plains
- 75 Southern Coastal Plain

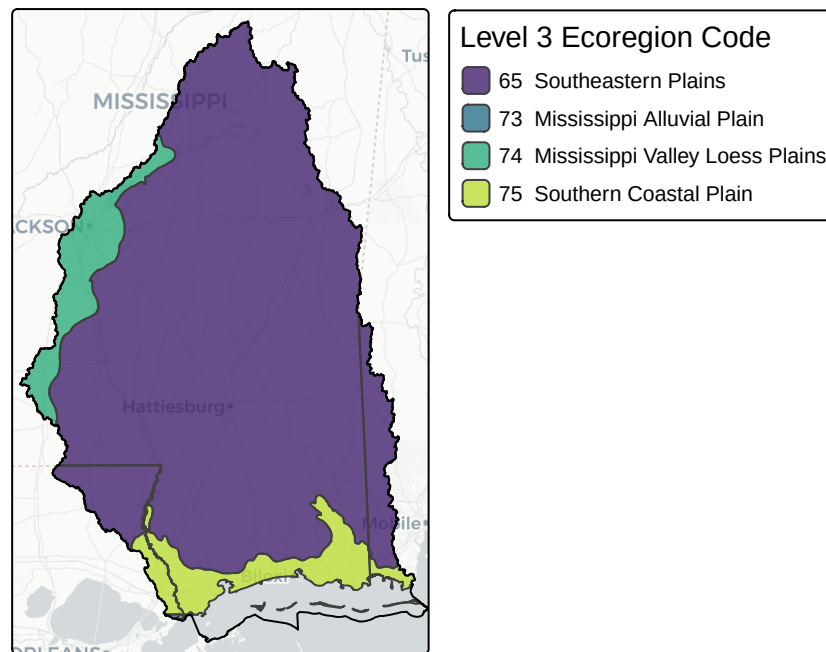


Figure 2.2: Level 3 Ecoregions in the Mississippi Sound Watershed only.

## 2.2 Level 4 Ecoregions

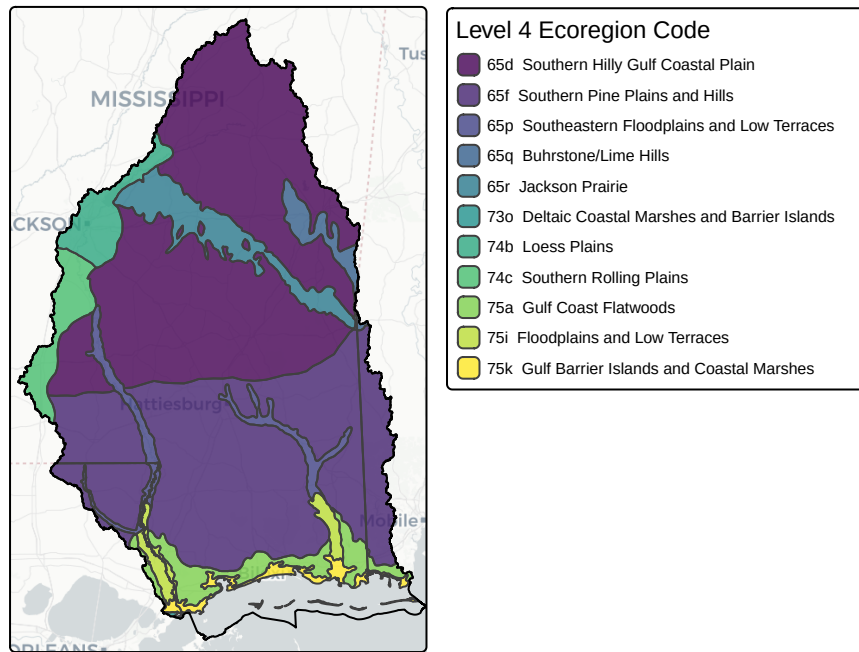


Figure 2.3: Level 4 Ecoregions in the Mississippi Sound Watershed.

### About

The `.qmd` file that generated this section was: **EPA\_\_Ecoregions.qmd**.

For information on data processing, see Appendix [A.1.2](#).

# **Part II**

## **Climatology**



Pieces of this part will be things like precipitation, temperature, etc.

## 3 Precipitation

The maps and information below were generated from gridded (spatial) climate normals: values averaged over 15, 30, or 100 years. Data were obtained from the National Centers for Environmental Information (NCEI).

### 3.1 Annual Totals

The coast receives more rain annually than more inland portions of Louisiana, Mississippi, and Alabama (Figure [3.1](#)). Based on 30-year climate normals (1991-2020), median annual precipitation in the Mississippi Sound Watershed is 60 inches. The lowest annual totals, around 56 inches per year, occur in the northern part of the watershed, with coastal areas receiving around 66 inches per year. These 30-year normals are somewhat higher than the 100-year baseline period of 1901-2000 (min: 52 in; median: 58 in; max: 64 in).

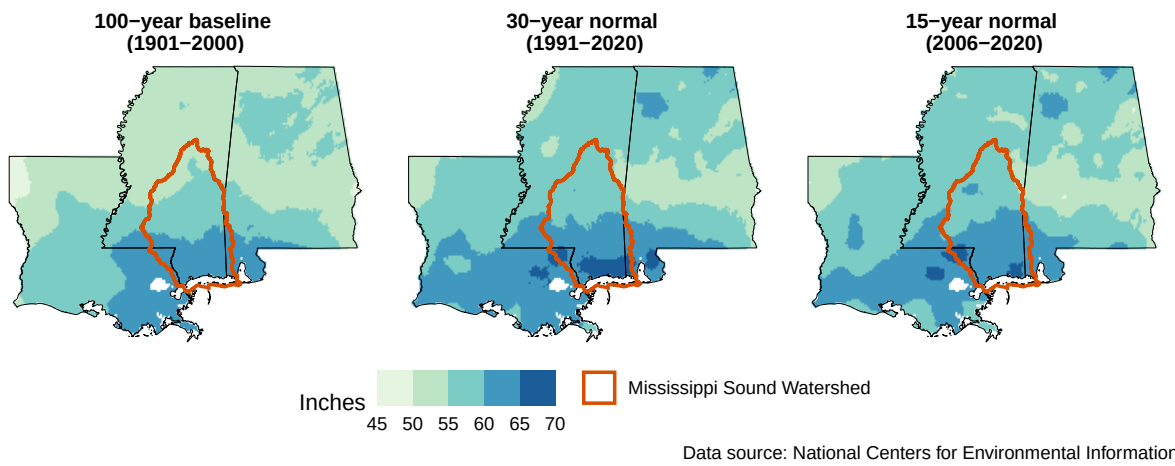


Figure 3.1: Annual Precipitation averaged over 100, 30, and 15 years.

## 3.2 Monthly Totals (30-year normal)

Seasonal patterns of precipitation also differ throughout the Mississippi Sound Watershed (Figure 3.2). Coastal areas receive most annual precipitation from June through August (up to 8 inches along the coast in July), while more inland areas receive the most from winter through spring. Tropical storms and hurricanes can and do cause higher precipitation totals than these averages; the peak of hurricane season is typically September-October.

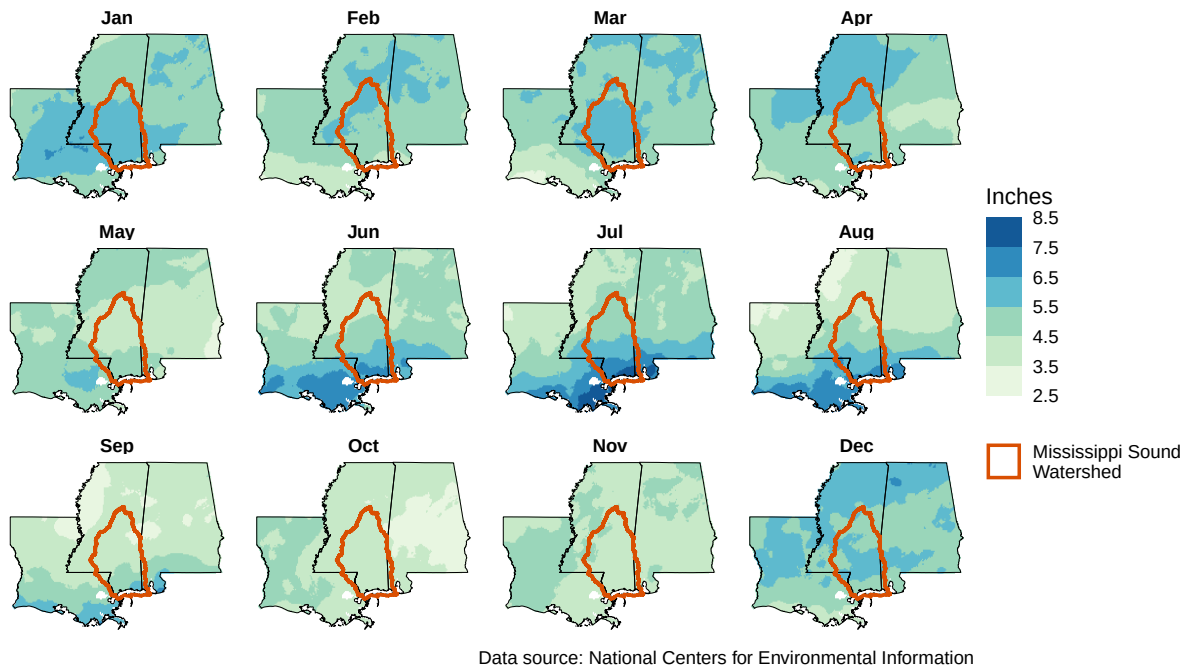


Figure 3.2: Monthly Precipitation, averaged over 30 years.

## About

The `.qmd` file that generated this section was: `precipitation.qmd`.

For information on data processing, see Appendix [A.2.1](#).

## 4 Temperature

The maps and information below were generated from gridded (spatial) climate normals: values averaged over 15, 30, or 100 years. Data were obtained from the National Centers for Environmental Information (NCEI).

### 4.1 Annual Mean Temperature

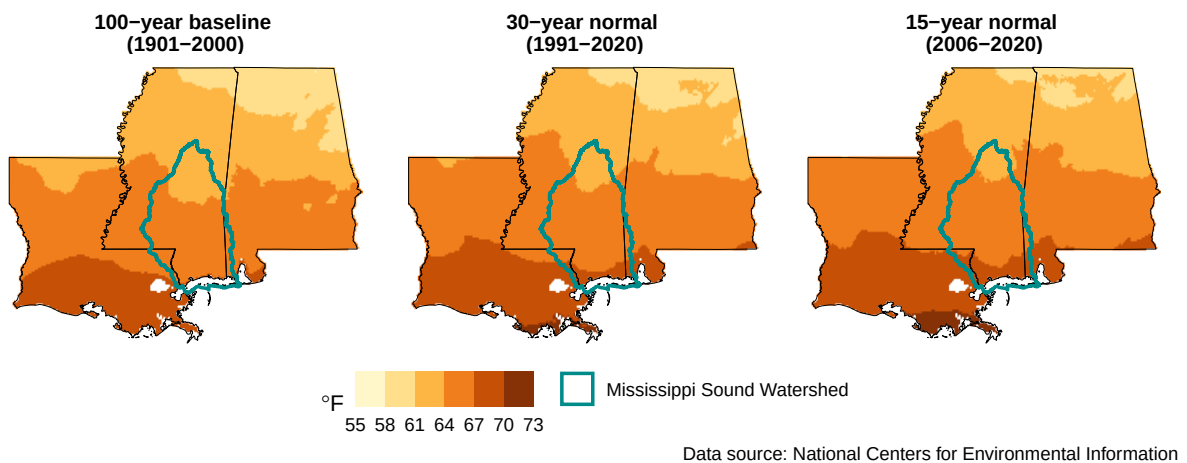


Figure 4.1: Annual Mean Temperature averaged over 100, 30, and 15 years.

## 4.2 Monthly Mean Temperature

### 4.2.1 30-year normals

Based on monthly averages over a 30-year period (1991-2020; see Table 4.1 and Figure 4.2), the coldest month in the Mississippi Sound Watershed is January. The lowest mean January value, in the northern part of the watershed, is 43.5 °F, with temperatures along the coast being higher at 52 °F. The hottest month is July, at 79.8 °F in the northern part of the watershed and 82.7 °F along the coast.

Table 4.1: Monthly mean temperature by month in the Mississippi Sound Watershed, averaged over 30 years (1991-2020). For each month, the minimum, mean, and maximum of the monthly means within the watershed are provided.

| Month | min  | mean | max  |
|-------|------|------|------|
| Jan   | 43.5 | 48.1 | 52.0 |
| Feb   | 47.4 | 52.0 | 55.7 |
| Mar   | 54.7 | 58.5 | 61.4 |
| Apr   | 61.9 | 64.9 | 67.9 |
| May   | 70.1 | 72.5 | 75.3 |
| Jun   | 77.2 | 78.8 | 81.1 |
| Jul   | 79.8 | 81.1 | 82.7 |
| Aug   | 79.3 | 80.6 | 82.9 |
| Sep   | 74.0 | 76.2 | 79.4 |
| Oct   | 63.1 | 66.1 | 70.8 |
| Nov   | 52.3 | 55.9 | 60.6 |
| Dec   | 45.8 | 50.1 | 54.3 |

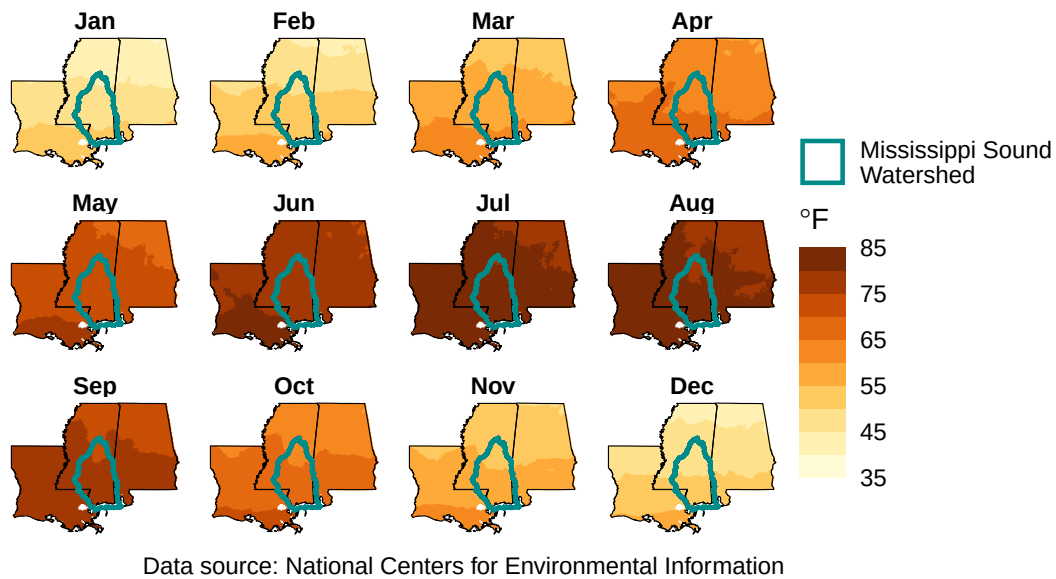
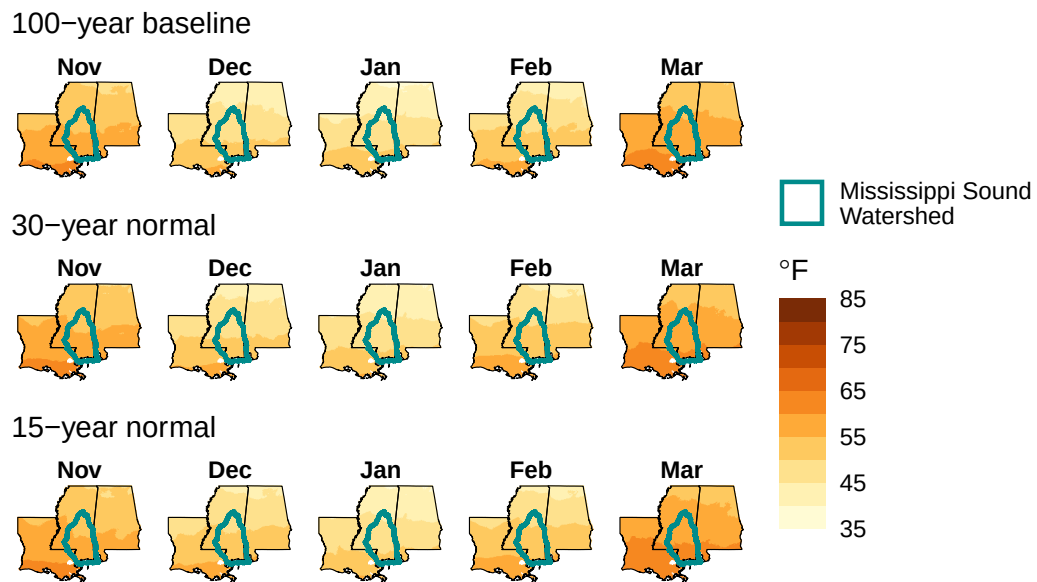


Figure 4.2: Monthly mean temperature averaged over 30 years (1991-2020).

#### 4.2.2 Comparing time periods

The maps below show only a subset of months. Each map contains the averages over a 100-year baseline, and 30- and 15- year normals. Each time period is a row, and the months line up vertically. Louisiana in particular shows some differences based on the time of aggregation in March and June.

#### 4.2.2.1 Cool Months



Data source: National Centers for Environmental Information

Figure 4.3: Comparison of long-term winter temperature averages across a 100-year baseline, and 30- and 15-year normals.



#### 4.2.2.2 Warm Months

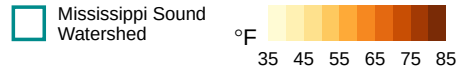
100-year baseline



30-year normal



15-year normal



Data source: National Centers for Environmental Information

Figure 4.4: Comparison of long-term spring and summer temperature averages across a 100-year baseline, and 30- and 15-year normals.

## About

The `.qmd` file that generated this section was: `temperature.qmd`.

For information on data processing, see Appendix [A.2.2](#).

## **Part III**

# **Hydrology**

Freshwater inflows from the different sources, salinity characteristics, etc.

## 5 Streams and Waterbodies

### 5.1 Amount of water

e.g. how many miles of streams, how many acres of lakes/ponds/etc. in the watershed

**TO-DO: Calculate this by basin (MDEQ definitions, not HUCs)**

#### 5.1.1 Stream Lengths

Table 5.1: Length (mi) of stream types in the MSEP watershed

| Type                       | total_miles |
|----------------------------|-------------|
| Artificial Path            | 1,496.8     |
| Canal/Ditch                | 145.3       |
| Coastline                  | 308.7       |
| Connector                  | 26.6        |
| Stream/River               | 5.2         |
| Stream/River: Intermittent | 20,350.3    |
| Stream/River: Perennial    | 11,730.7    |

#### 5.1.2 Waterbody Areas

Table 5.2: Total area (acres) of water body types in the MSEP watershed

| Type                                          | total_acres |
|-----------------------------------------------|-------------|
| Lake/Pond: Intermittent                       | 70.6        |
| Lake/Pond: Perennial                          | 67,602.0    |
| Lake/Pond: Perennial; Average Water Elevation | 27,726.5    |
| Lake/Pond: Perennial; Normal Pool             | 9.5         |
| Reservoir                                     | 48.2        |
| Reservoir: Aquaculture                        | 471.6       |
| Reservoir; Treatment                          | 954.8       |

Table 5.2: Total area (acres) of water body types in the MSEP watershed

| Type        | total_acres |
|-------------|-------------|
| Swamp/Marsh | 251,495.2   |

## 5.2 Designated Uses

The Mississippi Department of Environmental Quality (MDEQ), per the federal Clean Water Act of 1972, is responsible for classifying surface waters in Mississippi into ‘designated uses’ (e.g. fish and wildlife, recreation, shellfish harvesting, public water supply) and creating water quality standards to ensure designated uses can be met. More detail on designated uses and water quality standards can be found on [MDEQ’s website](#).

A 305(b) report (the most recent as of this writing is from 2024; [pdf available here](#)) is generated every other year, and quantifies how many and which water bodies are / are not meeting the criteria for their designated use.

Here, we are only summarizing designated uses within the Mississippi Sound Watershed - not assessment of whether criteria are met. See the 305(b) report linked above for more detailed information.

**The tables below do not include the general ‘Fish and Wildlife’ designation; that designation applies to most waters of the state.**

Table 5.3: Miles of streams with each designated use in the MS Sound Watershed

| Code    | Description                       | Miles   |
|---------|-----------------------------------|---------|
| REC     | Recreation                        | 1,011.6 |
| PWS     | Public Water Supply               | 26.3    |
| SHF-REC | Shellfish Harvesting & Recreation | 25.3    |
| FW1     | Fish and Wildlife                 | 10.4    |
| EPH     | Ephemeral                         | 1.8     |

Table 5.4: Area of non-linear waterbodies with each designated use in the MS Sound Watershed

| Code    | Description                       | Square Miles | Acres     |
|---------|-----------------------------------|--------------|-----------|
| REC     | Recreation                        | 742.2        | 475,019.3 |
| PWS     | Public Water Supply               | 34.8         | 22,259.2  |
| SHF-REC | Shellfish Harvesting & Recreation | 22.0         | 14,100.5  |
| PWS-REC | Public Water Supply & Recreation  | 0.2          | 119.8     |

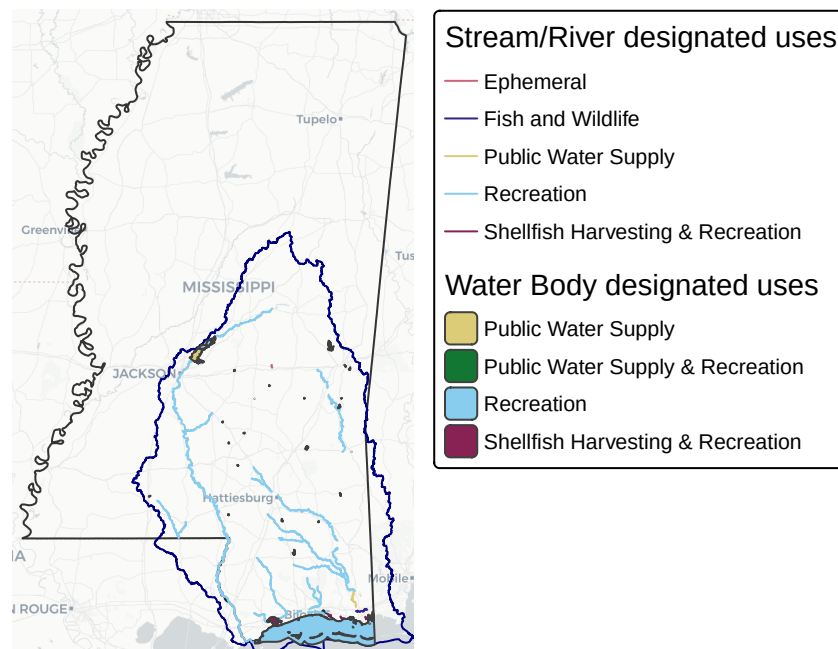


Figure 5.1: Designated uses of streams and waterbodies of the Mississippi Sound Watershed (navy outline on map), within the state of MS, as defined by MDEQ in the 2015 Water Quality Standards dataset. Data provided by MDEQ.

## 5.3 Impairments

303d, TMDLs - How many, where are they, what are the TMDLs for. What problems are we dealing with in the watershed.

## About

The `.qmd` file that generated this section was: `streams_waterbodies.qmd`.

For information on data processing, see Appendix [A.3.1](#).

## 6 Freshwater Inflows

Pearl River, Pascagoula River, bays. Do some summarizing based on USGS gage data?

### About

The `.qmd` file that generated this section was: `freshwater__inflows.qmd`.

For information on data processing, see Appendix [A.3.2](#).

## 7 Salinity

Summarize the info from the Gulf Data Atlas (e.g. the maps that went into the CCMP). Would be great to summarize the USGS salinity data and DMR's sampling data as well - some seasonal and annual details would be great.

### About

The `.qmd` file that generated this section was: `salinity.qmd`.

For information on data processing, see Appendix [A.3.3](#).



**Part IV**

**People and Land Use**

Here's where population calculations and land use/land cover summaries can be generated.

## 8 Population

The values calculated below are based on 2020 US Census data. The population estimate for the entire state of Mississippi was ~2.96 million in 2020.

### 8.1 Entire watershed

This estimate includes the portions of the watershed in Louisiana and Alabama:

**1,718,504**

### 8.2 Watershed in the state of MS

This estimate includes the state of Mississippi only:

**1,539,090**

### About

The `.qmd` file that generated this section was: **population.qmd**.

For information on data processing, see Appendix [A.4.1](#).

## 9 Land Use / Land Cover

The following information was generated using 2023 data from the [USGS Annual National Land Cover Database](#). See Section [A.4.2](#) for more detail.

### 9.1 Summary of Land Cover Types

### 9.2 Maps

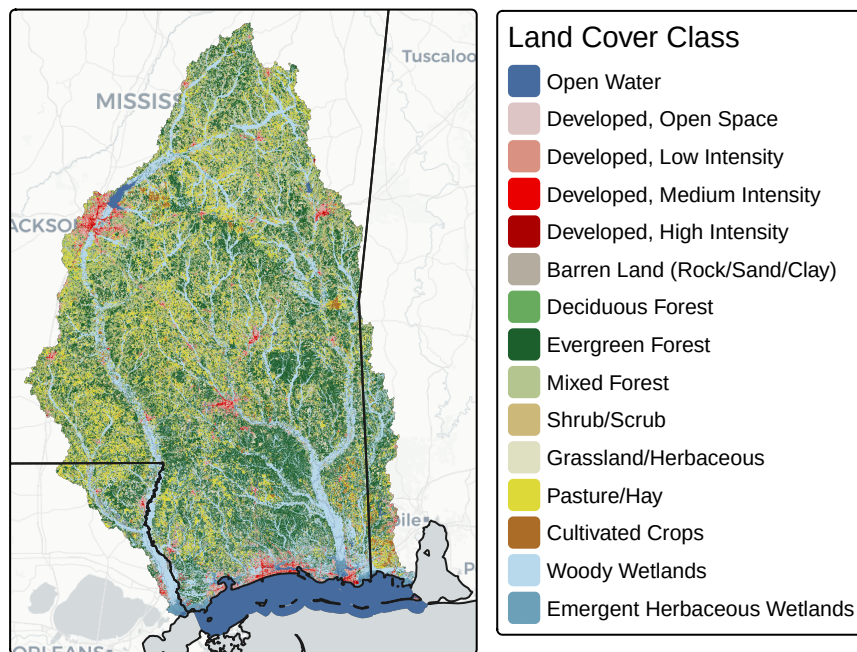


Figure 9.1: 2023 Land Cover designations from the USGS National Land Cover Database.

#### 9.2.1 Interactive

The pixels in this map have been aggregated so that 1 pixel represents each original 8x8 pixel grid (using the mode; the category that showed up most frequently in the grid). This was

| value | Land Cover Class             | Percent of Watershed |
|-------|------------------------------|----------------------|
| 11    | Open Water                   | 6.4                  |
| 21    | Developed, Open Space        | 5.8                  |
| 22    | Developed, Low Intensity     | 2.5                  |
| 23    | Developed, Medium Intensity  | 0.8                  |
| 24    | Developed, High Intensity    | 0.2                  |
| 31    | Barren Land (Rock/Sand/Clay) | 0.4                  |
| 41    | Deciduous Forest             | 1.0                  |
| 42    | Evergreen Forest             | 29.6                 |
| 43    | Mixed Forest                 | 8.8                  |
| 52    | Shrub/Scrub                  | 4.2                  |
| 71    | Grassland/Herbaceous         | 4.1                  |
| 81    | Pasture/Hay                  | 13.4                 |
| 82    | Cultivated Crops             | 0.8                  |
| 90    | Woody Wetlands               | 20.8                 |
| 95    | Emergent Herbaceous Wetlands | 1.3                  |

done to make it small enough to map interactively.

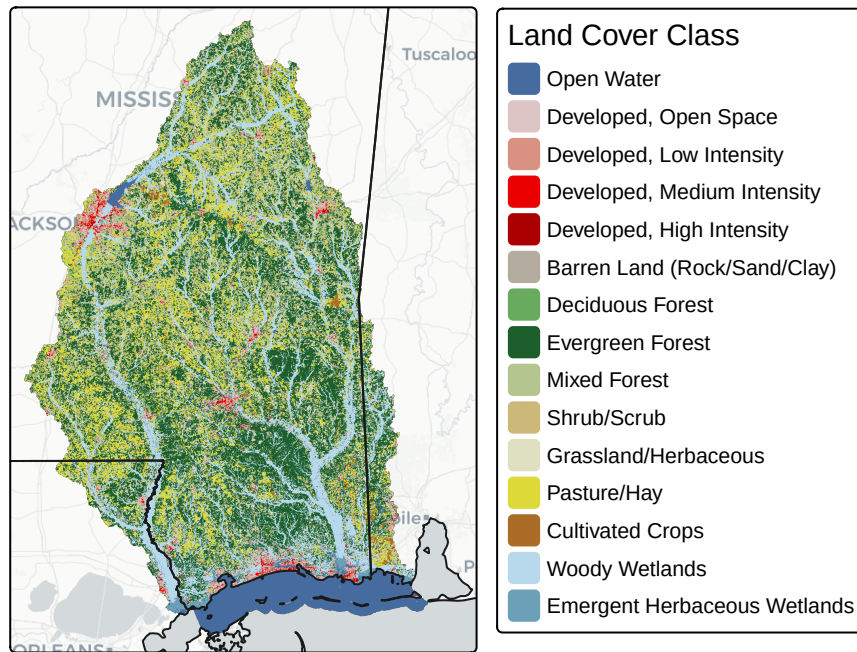


Figure 9.2: 2023 Land Cover designations from the USGS National Land Cover Database. Interactive when using the html format of this book.

## About

The `.qmd` file that generated this section was: `landuse_landcover.qmd`.

For information on data processing, see Appendix [A.4.2](#).

# Summary

Unsure what would go in a summary chapter .....

# A Data Processing

In general, links to original data sources are provided in each section below. Because data files are routinely updated, the files as downloaded and used in this document are also available online. These files are provided for reproducibility and transparency; for your own analyses, we encourage you to seek out data from the original linked sources.

## A.1 Geographic Setting

### A.1.1 Watersheds

*Jump to [Chapter 1](#)*

#### A.1.1.1 HUCs

stuff

#### A.1.1.2 MDEQ Basins

Shapefiles were provided by the Mississippi Department of Environmental Quality in August 2025. The zipped shapefiles are available in the [DEQ\\_basins](#) folder of the [msepCharacterization data files](#) directory. **However, we encourage you to contact MDEQ to get the most up-to-date files for your own analyses.**

### A.1.2 EPA Ecoregions

*Jump to [Chapter 2](#)*

Level 3 and 4 Ecoregion shapefiles were downloaded from the [EPA's website](#) on 16 June 2025 for the states of MS, AL, and LA.

The original downloaded files along with their metadata files are available in the [EPA EcoRegions](#) folder of the [msepCharacterization data files](#) directory.



Those files were read in, combined, and clipped to the Mississippi Sound Watershed boundaries (using the `outline_full` object from the [{msepBoundaries}](#) package) using the script `EPAEcoRegion_subsetting.R` ([on github](#)) in the `R_preprocessing` folder. Three objects resulted: `level3_lamsal`, level 3 ecoregions for the 3 states combined; `level3_mssoundwatershed`, level 3 ecoregions trimmed to only the Mississippi Sound Watershed area; and `level4_mssoundwatershed`, level 4 ecoregions trimmed to only the Mississippi Sound Watershed area. The level 4 ecoregions for the entire three states were not retained because there were so many categories; it really only makes sense to look at these at the smaller level.

The resulting objects were saved as layers of a geopackage, `EPA_EcoRegion.gpkg` in the `data/processed` folder. These files are available in the [processed](#) folder of the [msepCharacterization data files](#) directory.

## A.2 Climatology

### A.2.1 Precipitation

*Jump to Chapter 3*

Precipitation normals (over 15 and 30 years, and a 100-year baseline: 2006-2020, 1991-2020, and 1901-2000 respectively) were downloaded as “gridded normals” from the [National Centers for Environmental Information Climate Normals website](#) as netCDF (.nc) files. The downloaded files as used are available in the [Precipitation\\_NCEI](#) folder of the [msepCharacterization data files](#) directory.

These were subsetted to only monthly and annual value layers (removing layers with sd, min, max, and flags) and trimmed geographically to the states of MS, LA, and AL in the file `Precipitation_subsetting.R` ([on github](#)) in the `R_preprocessing` folder.

The resulting files were saved as netCDF files `prcp_15yearNormals.nc`, `prcp_30yearNormals.nc`, and `prcp_100yearBaseline.nc` in the `data/processed` folder. These files are available in the [processed](#) folder of the [msepCharacterization data files](#) directory.

### A.2.2 Temperature

*Jump to Chapter 4*

Average Temperature normals (over 15 and 30 years, and a 100-year baseline: 2006-2020, 1991-2020, and 1901-2000 respectively) were downloaded as “gridded normals” from the [National Centers for Environmental Information Climate Normals website](#) as netCDF (.nc) files. The downloaded files as used are available in the [Temperature\\_NCEI](#) folder of the [msepCharacterization data files](#) directory.

These were subsetted to only monthly and annual value layers (removing layers with sd, min, max, and flags) and trimmed geographically to the states of MS, LA, and AL in the file `Temperature_subsetting.R` ([on github](#)) in the `R_preprocessing` folder.

The resulting files were saved as netCDF files `tempAvg_15yearNormals.nc`, `tempAvg_30yearNormals.nc`, and `tempAvg_100yearBaseline.nc` in the `data/processed` folder. These files are available in the [processed](#) folder of the [msepCharacterization data files](#) directory.

## A.3 Hydrology

### A.3.1 Streams and Waterbodies

*Jump to Chapter 5*

#### A.3.1.1 Stream Lengths and Waterbody Areas

Stream Lengths and Waterbody Areas were calculated from the [EPA's NHDPlus](#) dataset. Files for the [South Atlantic \(03f\)](#) and [Lower Mississippi River \(03g\)](#) regions were downloaded on 12 June 2025. The original downloaded files are available in the [EPA NHDPlus](#) folder of the [msepCharacterization data files](#) directory.

Those files were read in, combined, and clipped to only the Mississippi Sound Watershed boundaries (using the `outline_full` object from the [{msepBoundaries}](#) package) using the script `EPANHD_subsetting.R` ([on github](#)) in the `R_preprocessing` folder. The resulting files were saved as geopackages `EPA_NHDplus_flowline_MSEP.gpkg` and `EPA_NHDplus_waterbody_MSEP.gpkg` in the `data/processed` folder. These files are available in the [processed](#) folder of the [msepCharacterization data files](#) directory.

#### A.3.1.2 Designated Uses

Shapefiles of designated uses, already limited to the portion of the Mississippi Sound Watershed that lies within the state of MS, were provided by the Mississippi Department of Environmental Quality (MDEQ) in April 2025. These files provide the designated uses for each waterbody from the 2015 Water Quality Standards dataset. The Mississippi Sound Watershed is made up of MDEQ's Pearl River, Pascagoula River, and Coastal Streams basins. Per MDEQ, "all waters in those 3 basins not specifically listed in the data provided are classified as Fish and Wildlife." See [their website](#) for more information.

The zipped shapefiles are available in the [DEQ\\_designated\\_uses](#) folder of the [msepCharacterization data files](#) directory. **However, we encourage you to contact MDEQ to get the most up-to-date files for your own analyses.**

### **A.3.1.3 Impairments**

Not done yet but see MDEQ's online [TMDL Tool](#).

### **A.3.2 Freshwater Inflows**

*Jump to [Chapter 6](#)*

Stuff

### **A.3.3 Salinity**

*Jump to [Chapter 7](#)*

Stuff - Gulf Data Atlas especially

## **A.4 People and Land Use**

### **A.4.1 Population**

*Jump to [Chapter 8](#)*

#### **A.4.1.1 Dasymetric Population Estimates**

The [EPA EnviroAtlas](#)'s dasymetric population datasets “intelligently reallocate ... population from census blocks to 30 meter pixels based on land cover and slope”. The national datasets for 2010 and 2020 were downloaded as raster files on 24 July 2025. Only the 2020 dataset has been used here.

The original downloaded file (zipped; ~3.6 GB) is available in the [Population](#) folder of the [msepCharacterization data files](#) directory.

The dataset was trimmed to Mississippi Sound Watershed boundaries (using the `outline_full` object from the [{msepBoundaries}](#) package) using the script `Population_2020_subsetting.R` ([on github](#)) in the `R_preprocessing` folder.

The resulting file, `Population_Dasymetric_2020.tif`, is available in the [processed](#) folder of the [msepCharacterization data files](#) directory.

## A.4.2 Land Use / Land Cover

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Land Use/Land Cover data was downloaded from the [USGS Annual National Land Cover Database](#) in April 2025. Each pixel represents 30x30m.

The original downloaded file, `Annual_NLCD_LndCov_2023_CU_C1V0.tif` (~1.4 GB), covers the entire continental US and is available in the [NLCD](#) folder of the [msepCharacterization data files](#) directory.

The dataset was trimmed to Mississippi Sound Watershed boundaries (using the `outline_full` object from the [{msepBoundaries}](#) package) using the script `NLCD_subsetting.R` ([on github](#)) in the `R_preprocessing` folder. Factor levels associated with the NLCD categories were also associated during this process and are part of the auxiliary file that loads with the dataset in R.

The resulting files, `NLCD_MSEP.tif` and `NLCD_MSEP.tif.aux.xml` (the file with associated factor levels), are available in the [processed](#) folder of the [msepCharacterization data files](#) directory.

## About

The `.qmd` file that generated this section was: `data_processing.qmd`.